CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 11 : Python Program for Map Coloring to Implement CSP

Aim

To develop a Python program that colors the regions of a map so that no two adjacent regions have the same color, using a Constraint Satisfaction Problem (CSP) approach.

Algorithm

- Define the list of available colors and the map as a graph of neighbouring regions.
- Create an empty dictionary to store the color assigned to each region.
- For each region, try the colors one by one in order.
- Assign the first color that is not already used by any neighbouring region.
- Display the final color assignment of all regions.

Program

```
colors = ["Red", "Green", "Blue"]

graph = {
    "A": ["B", "C"],
    "B": ["A", "C"],
    "C": ["A", "B"]
}

color = {}

for node in graph:
    for c in colors:
        if all(color.get(neighbor) != c for neighbor in graph[node]):
            color[node] = c
            break

print(color)
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/11.py ==============
{'A': 'Red', 'B': 'Green', 'C': 'Blue'}
>>>
```

RESULT

Thus, the Map Coloring problem was solved successfully using CSP in Python.